FOURTH SEMESTER M.Sc. DEGREE EXAMINATION, JUNE 2019

(CUCSS—PG)

Chemistry

CH 4C 12—INSTRUMENTAL METHODS OF ANALYSIS

(2015 Admissions)

Time : Three Hours

Maximum : 36 Weightage

Section A

Answer **all** *questions.*

Each question carries a weightage of 1.

1. What is a determinate error ? How you ascertain whether it is constant or proportionate ?
2. What is meant by the term 'confidence limit' ?
3. Explain the term 'indicator range'.
4. Bring out the functioning of an adsorption indicator with a suitable example.
5. How do you determine the $[H^+]$ of an add solution using potentiometric method ?
6. Mention the advantages of using dropping mercury electrode in polarography.
7. What are advantages of coulorimetric titrations ?
8. List the factors that affect fluorometric analysis.
9. Explain the function of atomizer in AAS.
10. What informations do you gather from X-ray powder diffraction pattern ?
11. Why α-alumina is used as reference material in DTA ?
12. What are the factors that affect TG curves and how ?

(12 × 1 = 12 weightage)

Section B

Answer any **eight** *questions.*

Each question carries a weightage of 2.

13. A sample metal chloride on analysis showed the percentage of metal as :

    32.72, 32.78, 32.98, 32.84 and 32.79. Calculate : (a) Mean deviation ; (b) Standard deviation ; and (c) Co-efficient of variation.

14. Compare the method of averages and method of least squares for the treatment of analytical data.

15. Differentiate between co-precipitation and post-precipitation. How do they affect quantitative analysis ? How they can be avoided ?

16. Describe the use of oxine as a precipitant In gravimetric analysis.

17. Critically evaluate the graphical methods for determining the end point in potentiometric titrations.

18. Write a note on biosensors.

19. Write briefly on spectrophotometrlc titrations.

20. What is Auger effect ? How do you distinguish between XPS peaks and AES peaks in an electron spectrum ?

21. Discuss the use of SEM in surface analysis.

22. 'TG studies always require DTA also'. Comment on this statement.

23. Discuss the principle involved in isotope dilution methods.

24. What are the characteristics of an ideal detector of gas chromatography ?

(8 × 2 = 16 weightage)

Section C

Answer any two questions.
Each question carries a weightage of 4.

25. Discuss the principle involved in amperometric titrations ? Give an account of the different types of amperometric titrations. Mention their advantages.

26. Discuss the principle and applications of atomic absorption spectrometer. What are its merits and demerits ?

27. Discuss the principle, experimental set up and applications of neutron activation analysis.

28. Outline the principle, instrumental set up and applications of HPLC. What are the advantages of HPLC over other methods of chromatography ?

(2 × 4 = 8 weightage)